

Ahmed Toba Mahmoud

Software Engineer — Backend Developer

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Professional Summary

Software Engineer with a solid engineering foundation in computer science fundamentals, including data structures, algorithms, object-oriented programming, software design, and system architecture. Hands-on experience developing production-ready backend solutions and collaborating within agile teams to deliver reliable software. Passionate about building scalable, secure, and maintainable systems while applying engineering best practices throughout the software development lifecycle. Continuously driven to learn, solve challenging problems, and contribute to impactful software products.

Education

B.Sc. in Communications and Electronics Engineering | Faculty of Engineering, Beni Suef University

Grade: Very Good

2026

Graduation Project | Forging a Secure and Resilient Digital Backbone for a New Era of Academia

Grade: Excellent

[Github](#) | [Live Demo](#)

- Led a multidisciplinary engineering team in planning, architecting, and delivering a secure Smart University ecosystem spanning Backend Development, Cloud & DevOps, AI, IoT, Enterprise Networking, and Cybersecurity.
- Served as Technical Lead for the Backend, AI Chatbot, and IoT modules, designing the system architecture and leading the implementation of core platform components.
- Designed and implemented scalable backend services, RESTful APIs, real-time communication using Socket.IO, authentication and authorization mechanisms, and cloud-integrated system architecture following modern software engineering practices.
- Developed a cloud-connected biometric attendance system using ESP32, Azure IoT Hub, secure device communication, offline synchronization, and real-time attendance processing. [Github](#)
- Engineered a multi-layer AI academic assistant featuring general academic Q&A, role-based live data retrieval, and a RAG-powered document intelligence system, with token budget optimization and defenses against prompt injection through secure role-based data access.

Professional Experience

Backend Developer Intern | Hi Deep AI | Beni Suef, Egypt

Sep 2025 – Dec 2025

- Collaborated directly with clients across the full Software Development Lifecycle (SDLC), contributing to business analysis, system analysis, solution architecture, implementation, testing, and project delivery.
- Designed, developed and enhanced scalable backend services, RESTful APIs, MongoDB data models, file upload functionality, and core application features within an Agile development team.
- Contributed to system architecture, feature planning, containerized deployment using Docker, version control with Git/GitHub, and supported deployment workflows on AWS.
- Wrote unit tests, participated in code reviews, and resolved technical issues to improve software quality, reliability, and maintainability.

Projects

Natours – Full-Stack Tour Booking Platform

[GitHub](#) | *Node.js · Express.js · MongoDB · JWT · Stripe · Pug*

- Built a production-ready full-stack tour booking platform featuring secure RESTful APIs, server-side rendering (SSR), and responsive user experience.
- Implemented JWT authentication, Stripe payments, rate limiting, XSS/NoSQL protection, and other security best practices.

E-Commerce Platform – Full-Stack Shopping Platform

GitHub | Angular · Node.js · Express.js · MongoDB

- Developed a role-based full-stack e-commerce platform with product, cart, order, and authentication modules.
- Implemented secure backend APIs and integrated frontend workflows for a complete shopping experience.

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript, C++, C, SQL
- **Backend Development:** Node.js, Express.js, RESTful APIs, API Security, Socket.IO
- **Databases:** MongoDB, MySQL
- **Cloud & DevOps:** Microsoft Azure, Azure IoT Hub, Docker, AWS, Git, GitHub
- **AI Engineering:** RAG, Prompt Engineering, Prompt Injection Mitigation, Role-Based AI Systems, Token Budget Optimization, LLM Integration
- **IoT Development:** ESP32 Firmware Development, MQTT, Biometric Integration, IoT Device Communication
- **Core Fundamentals:** OOP, Data Structures & Algorithms, SOLID Principles, Software Design
- **Developer Tools:** Postman, Linux

Training & Certifications

- **MEAN Stack Web Development (120 Hours)** – National Telecommunication Institute (NTI) 2025
- **Node.js, Express, MongoDB & More: The Complete Bootcamp** – Udemy (Jonas Schmedtmann) 2025

Leadership & Volunteering

Co-Founder & Business Developer | Maymay Start-up

Jan 2025 – Jan 2026

- Co-led business strategy, product development, and partnership initiatives to support startup growth.

Rally BSU | Beni Suef University

HR Manager

Aug 2025 – Aug 2026

Led recruitment, team development, conflict resolution, and HR operations.

Marketing Vice

Feb 2025 – Aug 2025

Directed marketing initiatives, campaign planning, and cross-functional team coordination.

Head of HR | GDSC EELU

Feb 2024 – Aug 2024

Managed recruitment, onboarding, mentoring, and student progress throughout Google Developer Student Clubs learning programs.

Languages

- Arabic:** Native
- English:** Professional Working Proficiency